KUNAL MANOCHA

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Education

Dronacharya College of Engineering, Gurugram — **Aug 2023 – May 2027**

Bachelor of Technology in Computer Science with Artificial Intelligence & Machine Learning — CGPA *7.84/10 (till Sem VI)*

12th ICSE – *Lord Jesus Public School (2021-22)* — *87.6%*

10th ICSE – *Lord Jesus Public School (2019-20)* — *80.33%*

Technical Skills

Programming Languages: Python, Java, C++, C, SQL, Kotlin, Dart

Frontend & App Development: HTML5, CSS, JavaScript, ReactJS, NodeJS, Jetpack Compose, UI/UX Principles, REST APIs, XML, JSON

Frameworks & Libraries: PyTorch, Scikit-learn, OpenCV, Pandas, NumPy, Matplotlib

Machine Learning & AI: Supervised & Unsupervised Learning, Computer Vision, NLP, Artificial Intelligence, Machine Learning

Software Engineering & Core CS: Object Oriented Programming (OOP), Data Structures & Algorithms, Software Engineering.

Databases & Cloud Technologies: MySQL, MongoDB.

Tools & Version Control: Git, GitHub, Visual Studio, Python Scripting & Automation, Jupyter Notebook

Experience

Software Development Engineer Intern – The Garden Store Pvt. Ltd. — **June 2026 – July 2026**

Gurugram, India– Hybrid

- *Built a responsive **B2B e-commerce web application** using **React, Node.js** and **Express**, implementing product catalogs, cart workflows and order tracking.*
- ***Implemented secure authentication and authorization** using **OAuth 2.0 and JWT**, integrating role-based access control with **MongoDB**.*
- ***Optimized web performance** using **Redux Toolkit** for centralized state management and efficient client-side caching.*
- ***Followed MVC architecture and Git workflows**, collaborating on modular frontend components and backend services.*

Projects

MorphoMinds | AI-Powered Hyper-personalized Learning Platform for Dyslexic Children | Link

- *Designed and developed a **web platform** to help dyslexic children learn reading and writing without a teacher through **AI-based** visual and auditory cues.*
- *Implemented **interactive modules** to assist children in recognizing letters and words using **gesture and voice interaction**.*
- *Prioritized accessibility and **child-friendly UX** to improve engagement and independent learning.*

Park Prabandh | An Agentic-AI powered Smart Parking Automation System | Link

- *Developed a **responsive web-based** smart parking management platform using **ReactJS** and **TypeScript**, enabling real-time monitoring, slot visualization, and seamless user interaction.*
- *Developed an Agentic-AI system using **OpenCV** and **IoT** for autonomous vehicle entry and parking slot detection.*
- *Engineered a multi-layer architecture (Edge to Agentic) with autonomous workflows for dynamic slot allocation and integrated **FASTag-based** payments.*
- *Implemented a **3D analytics dashboard** to monitor occupancy and traffic patterns, minimizing human intervention and ensuring revenue protection.*

Research Work

(Ongoing) Quantum AI for Personalized Nutrition using Genomic, Metabolic and Lifestyle Data: A Hybrid Framework

- *Designed a hybrid quantum-classical architecture to integrate genomic, metabolic, and real-time lifestyle data.*
- *Applied Quantum Machine Learning (VQC) for high-dimensional feature mapping and analysis of multi-modal health datasets.*
- *Developed an adaptive recommendation framework utilizing continuous feedback loops for dynamic dietary adjustments.*

Certifications / Achievements

- Completed **290+** DSA Questions on **LeetCode**. ***LeetCode Profile Link***
- Scored **All India Rank 5** in **NCAT 2025**. ***Certificate***
- **Deviators Devolympus 2025 (DCE)** : Secured **1st-runner up** place in intra-college hackathon for creating **MorphoMinds** – An AI Powered Learning Platform for Dyslexic children.
- **NPTEL** : Secured **Bronze** in **Programming in Java** certification.
- Generative-AI for Software Development certification by **IBM SkillsBuild**.
- **Infosys Springboard** certifications: **Basics of Python**, **Introduction to Data Science**, **Data Structures & Algorithms**.
- Completed the **Japanese Language (N5) certification course** from **Udemy.** ***Certificate***